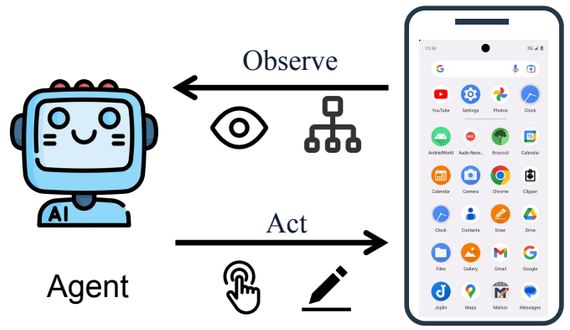
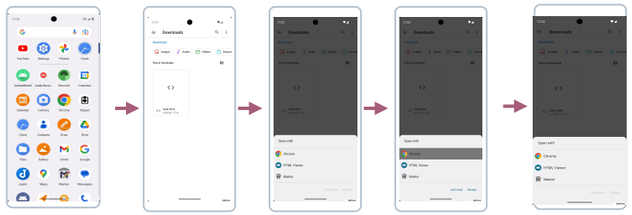
System Setup & Execution

Task Execution Pipeline



Execution Traces



Screenshots GUI Actions Model logs UI hierarchy

Agent Implementations

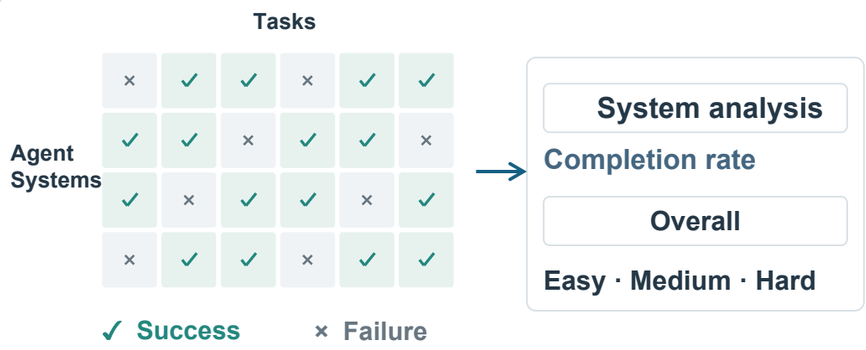
- P1 Visual ReAct
- P2 XML-Assisted ReAct
- P3 Planner–Executor–Reflector
- P4 Hierarchical Reflection

Foundation Models

Qwen · Gemini · ChatGPT

Stage 1: Autonomous Task Completion RQ1

Task outcomes



Task analysis

- Solved completion
- System dependent
- Unsolved tasks

Stage 2: Failure Causes RQ2

1 Failure Taxonomy Construction

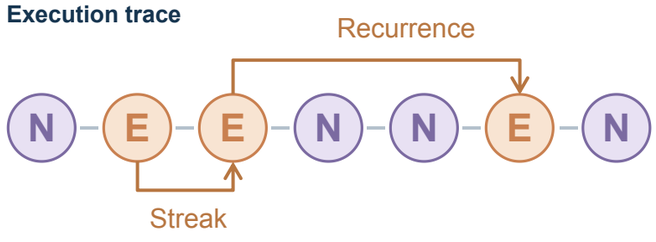


2 Failure Taxonomy



Stage 3: Execution Dynamics RQ3

1 Error Dynamics (Temporal Evolution)



Analyzed aspects

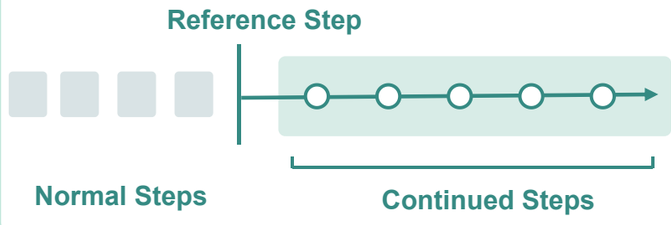
- **Error proportion** — How often errors occur
- **Error streak** — Consecutive error length
- **Error recurrence** — Whether errors reappear

2 GUI State Factors (Static GUI Indicators)

- Screen Similarity
- Actionable Element Area
- Spatial Crowding
- Edge Density

Stage 4: Continued Execution RQ4

Continuation Setup



Completion Yield & Resource Cost

- **Completion gain** — Increase in completion rate
- **Runtime** — Additional execution time
- **Token usage** — Additional token consumption